

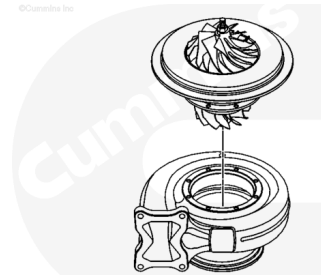
Trainings ▾

Assemble

⚠ CAUTION ⚠

The turbine blades can be easily damaged when installing the assembly in the turbine housing.

Assemble the center housing rotating assembly on to the turbine housing.

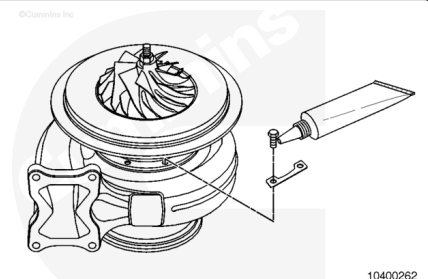


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Lubricate the threads of the new capscrews with anti-seize compound, Part Number 3824732, or equivalent.

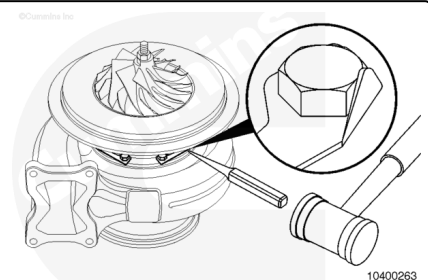
Install the capscrews, with new lock plates, to the turbocharger turbine housing.

Torque Value: 20.3 n·m [180 in-lb]



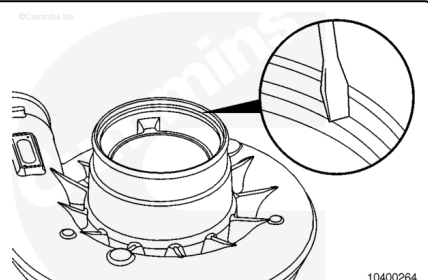
10400262

Use a hammer and punch to fold the locking plate onto the flat of the capscrew to prevent loosening.



10400263

Assemble the inlet baffle into the compressor cover and secure with a retaining ring. Hold one end of the retaining ring in the groove. Press the remainder of the retaining ring into position. Use a flat tip screwdriver to make sure the retaining ring is correctly seated in the compressor cover groove.

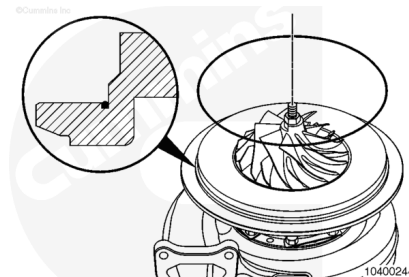


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Note : Turbocharger HX83 uses a titanium compressor wheel. The edge of the wheel is very sharp. Handle the

compressor wheel with care.

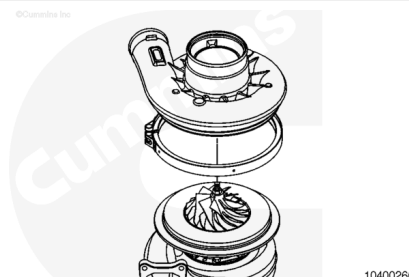
Assemble the compressor housing new o-ring on the turbocharger.



⚠ CAUTION ⚠

Do not damage the compressor wheel blades during assembly.

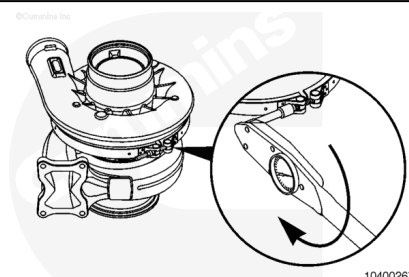
Assemble the v-band clamp and compressor cover onto the turbocharger.



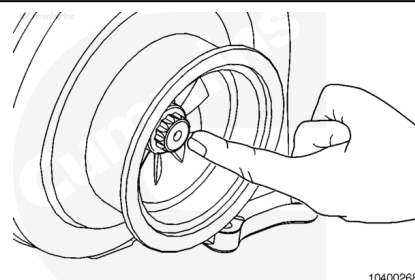
Place the v-band clamp in position.

Tighten the v-band clamp.

Torque Value: 12 n•m [106 in-lb]



Make sure the compressor wheel and turbine wheel rotate freely.

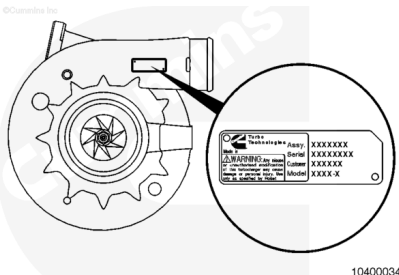


Install

All Applications Except Marine and Rear Mount Rail

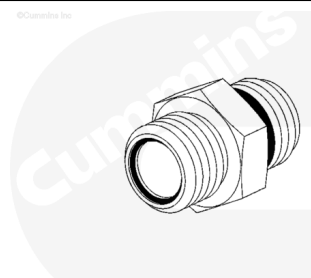
Cummins Turbo Technologies™ turbocharger(s) are used on the QSK19 engine. See the data tag on the turbocharger to determine the model.

The name is also cast on the housing of a turbocharger.



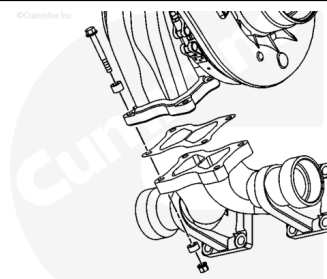
Note : All turbocharger fittings are the flat face o-ring type fittings. Make sure the o-ring is in place before attaching the hose(s) to the fittings.

The Cummins Turbo Technologies™ turbocharger uses a 9/16-18 UNF, straight thread, o-ring type oil supply fitting.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Apply an anti-seize compound to the mounting capscrews.

Install the gasket.

Note : The word **OUT** or raised gasket bead **must** be positioned toward the turbocharger.

For engines with mechanically and electronically actuated injectors, install the turbocharger, turbocharger capscrews, nuts, and spacers. The 12.7 mm [1/2 inch] spacer **must** be on the turbocharger side of the joint, while the 9.525 mm [3/8 inch] spacer **must** be on the manifold side or the bolted joint.

Tighten the turbocharger capscrews using sequence and torque steps below.

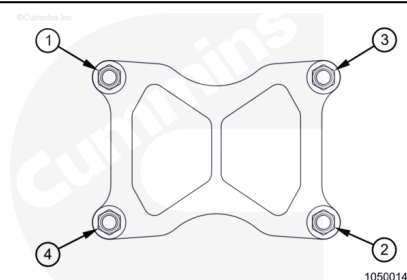
Torque Value:

Step 1: 20 n•m [177 in-lb]

Torque Value:

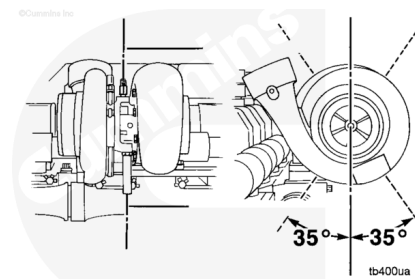
Step 2: 47 n•m [35 ft-lb]

Perform the final torque value in Step 2 using the sequence shown a second time.



Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

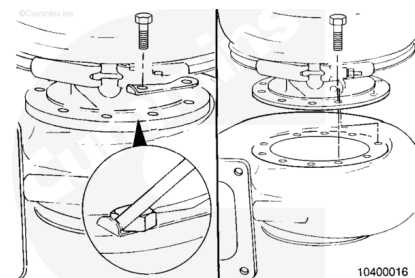
The turbocharger uses capscrews to attach the bearing housing to the turbine housing.



For bearing housing adjustment, bend the lockplate off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

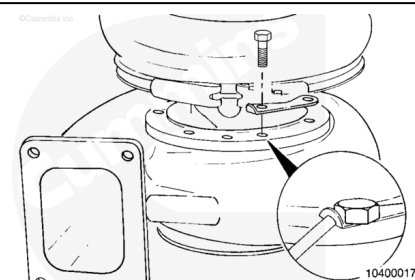


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 20 n•m [177 in-lb]

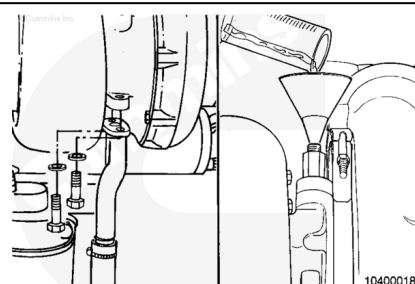
Bend the lockplate tabs over the capscrews.



Install the oil drain line. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)

Pour clean engine oil into the turbocharger oil supply fitting.

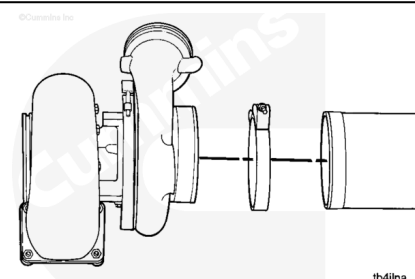
Turbocharger Oil Supply: 50 to 60 cc [2.0 to 3.0 oz.]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

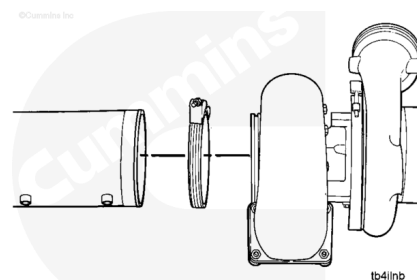
Torque Value: 8 n•m [71 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the v-band clamp.

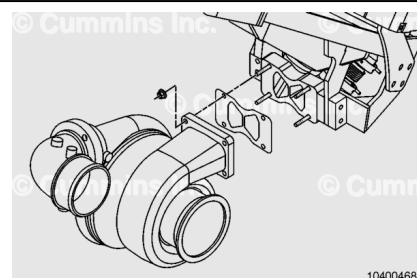
Torque Value: 8 n•m [71 in-lb]



Rear Mount Rail Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Apply an anti-seize compound to the mounting studs.

Install the gasket on the studs. The word **OUT** **must** be positioned toward the turbocharger.

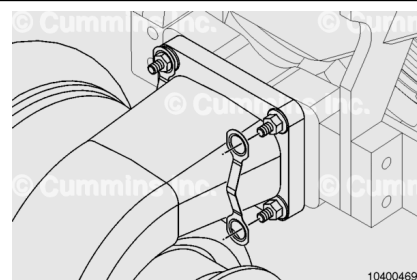
Install the turbocharger.

Install and finger tighten the nuts on the studs.

Tighten the mounting nuts in a criss-cross pattern.

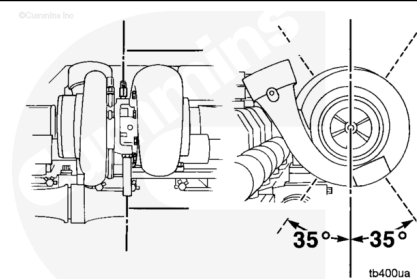
Torque Value: 57 n•m [42 ft-lb]

Install the two lockplates over the turbocharger mounting nuts.



Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

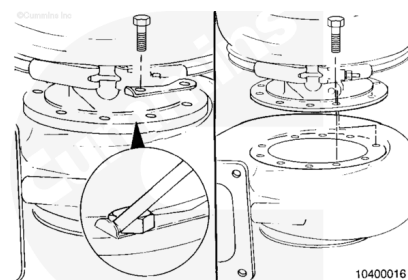
The turbocharger uses capscrews to attach the bearing housing to the turbine housing.



For bearing housing adjustment, bend the lockplate off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

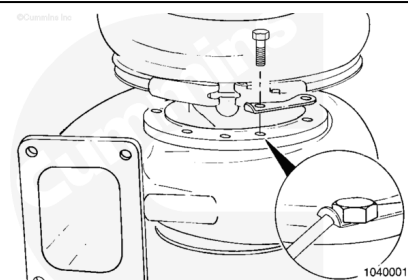


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 20 n•m [177 in-lb]

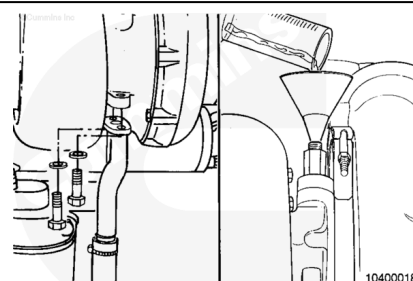
Bend the lockplate tabs over the capscrews.



Install the oil drain line. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)

Pour clean engine oil into the turbocharger oil supply fitting.

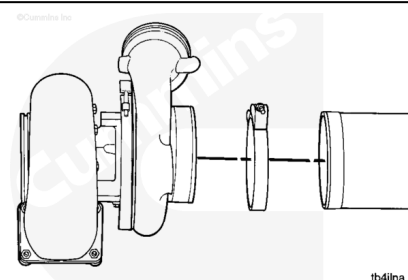
Turbocharger Oil Supply: 50 to 60 cc [2.0 to 3.0 oz.]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

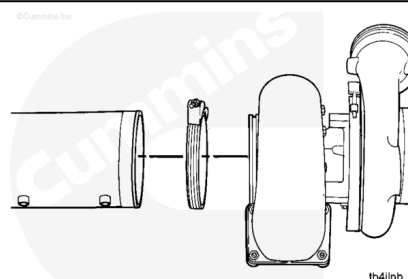
Torque Value: 8 n•m [71 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the v-band clamp.

Torque Value: 8 n•m [71 in-lb]



Last Modified: 28-Oct-2022